# CSE 311:
# Data Communication

**Instructor:**

**Dr. Md. Monirul Islam**

# Analysis of Signals and systems

## Topics

- Definition, classification and properties of signals
- Examples of some useful signals
- Fourier Series of Periodic Signals

# Signal Transmission through Linear Time Invariant System

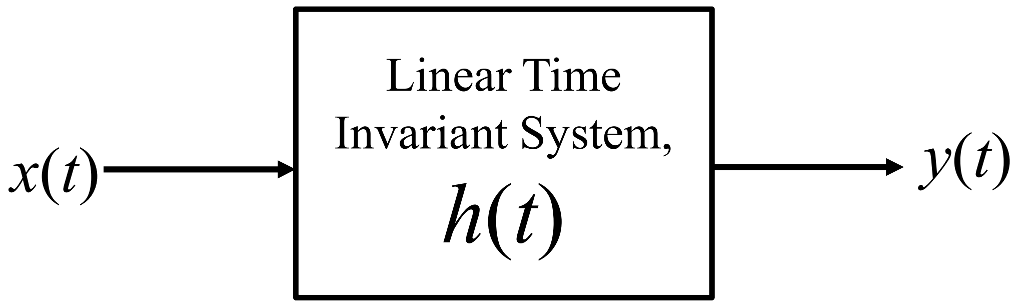
Linear Time Invariant System

$$\sum_k a_k \delta(t - t_k) \longrightarrow \boxed{\begin{array}{c}\text{Linear Time} \\ \text{Invariant System,} \\ h(t)\end{array}} \longrightarrow \sum_k a_k h(t - t_k)$$

LTI system is described by impulse response, $h(t)$

# Signal Transmission through Linear Time Invariant System

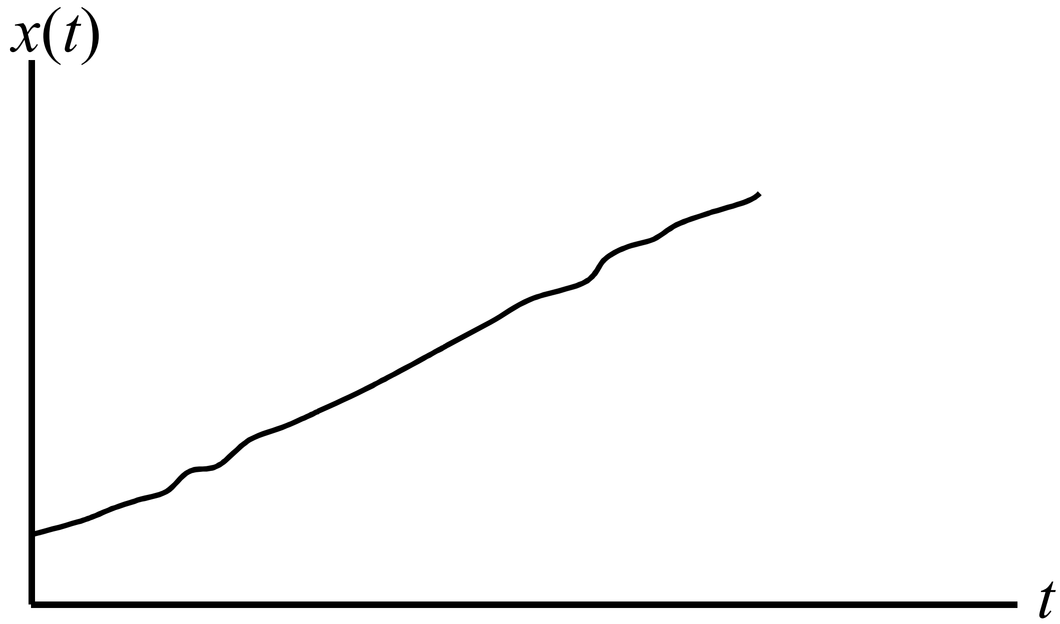
## Linear Time Invariant System




We will find relation betn $x(t)$ and $y(t)$

# Signal Transmission through Linear Time Invariant System

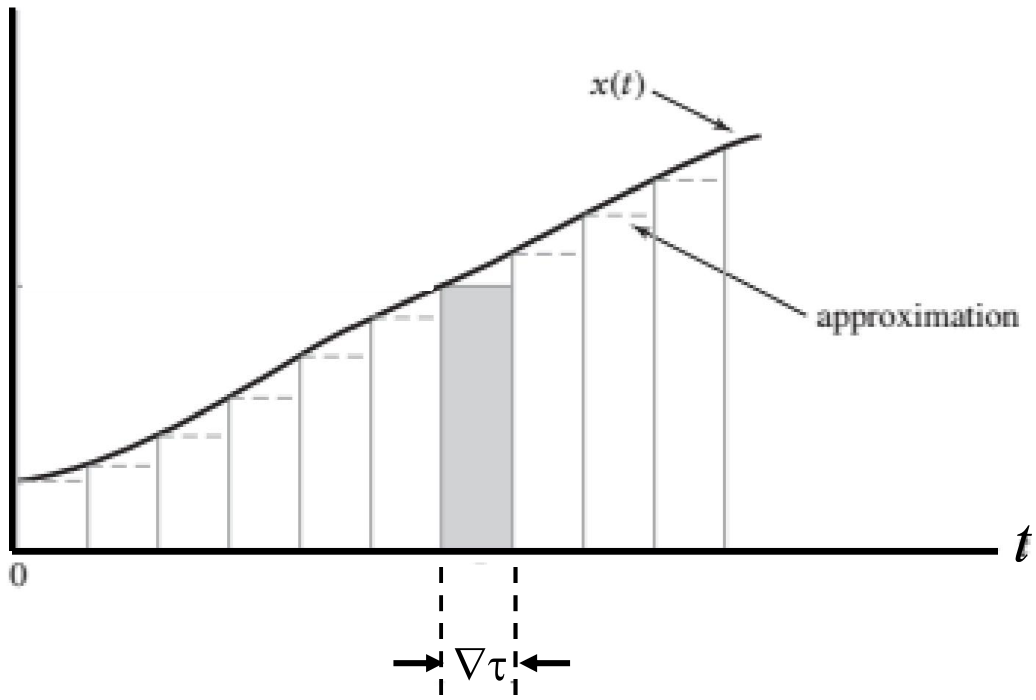
Linear Time Invariant System




Let the input signal be $x(t)$

# Signal Transmission through Linear Time Invariant System

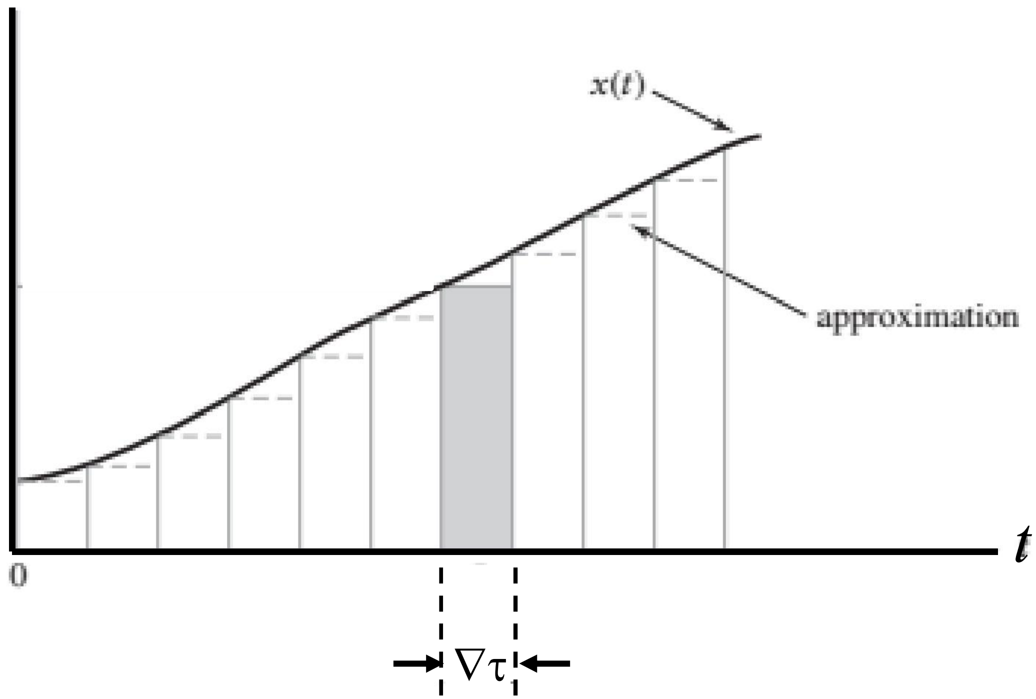
Linear Time Invariant System




Stair case approximation of $x(t)$ using narrow rectangular pulses

# Signal Transmission through Linear Time Invariant System

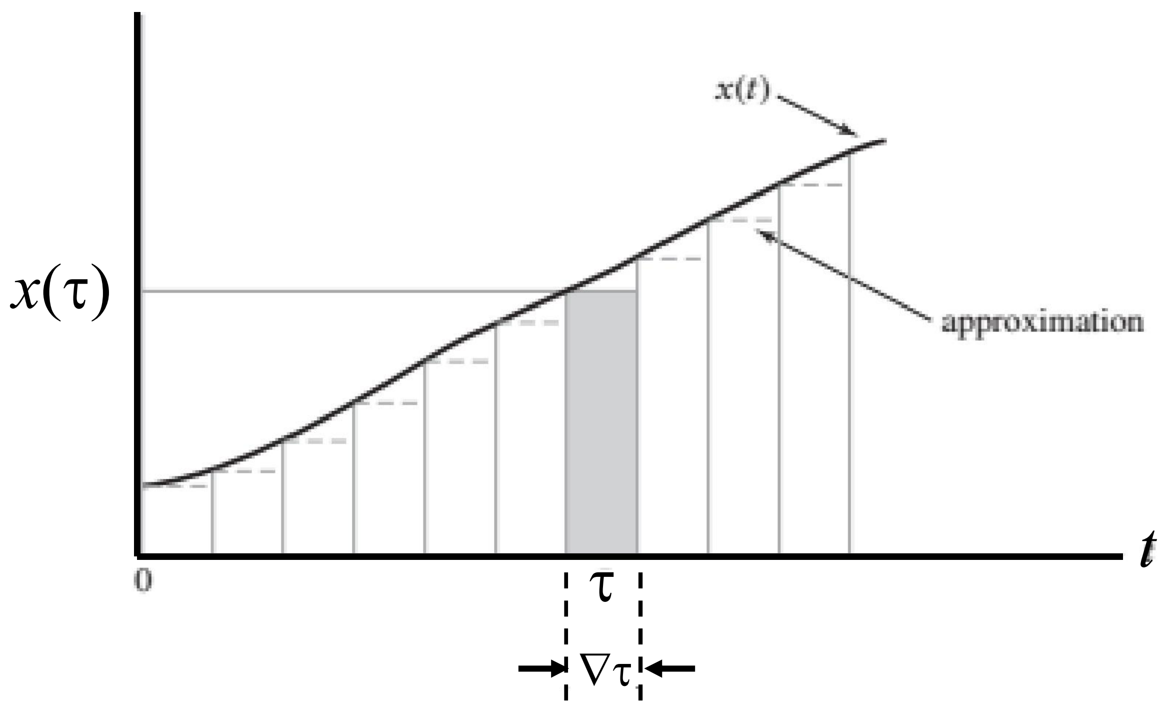
Linear Time Invariant System




As ∇τ approaches to 0, pulse approaches to δ(.) ×height× ∇τ

# Signal Transmission through Linear Time Invariant System

Linear Time Invariant System

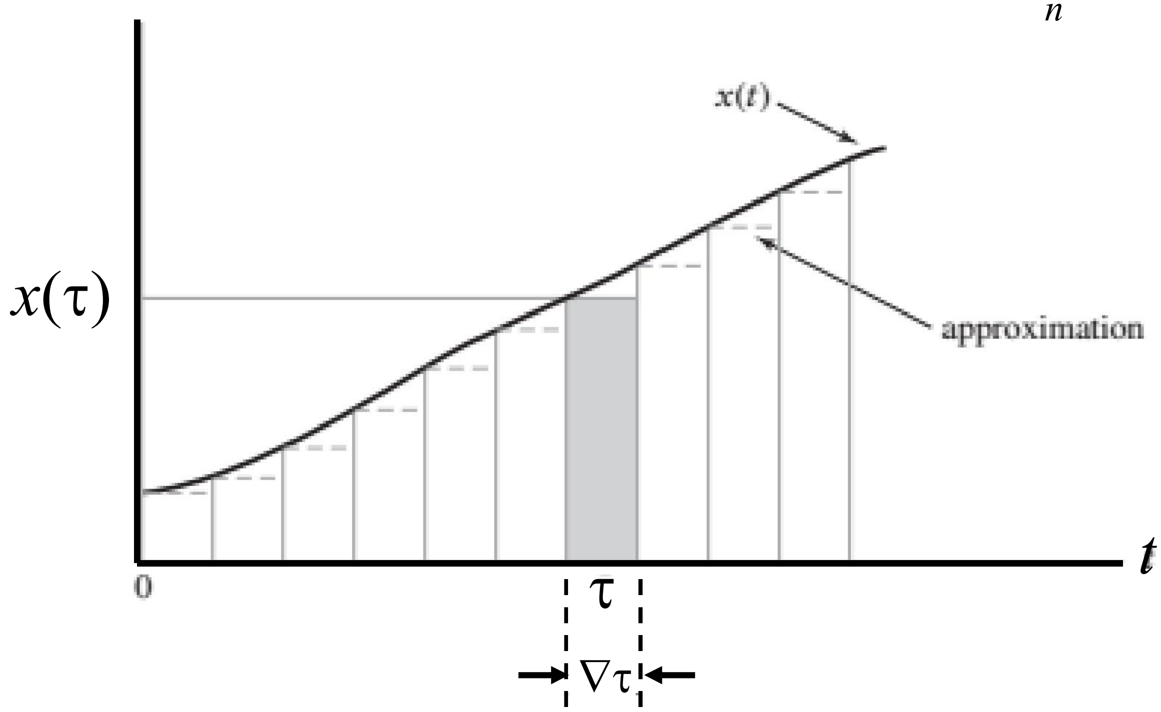



At $t = \tau$, pulse approaches to $\delta(t-\tau)x(\tau)\,\nabla\tau$

# Signal Transmission through Linear Time Invariant System

Linear Time Invariant System

$$x(t) = \sum_{n} \delta(t - n\nabla\tau)x(n\nabla\tau)\nabla\tau$$



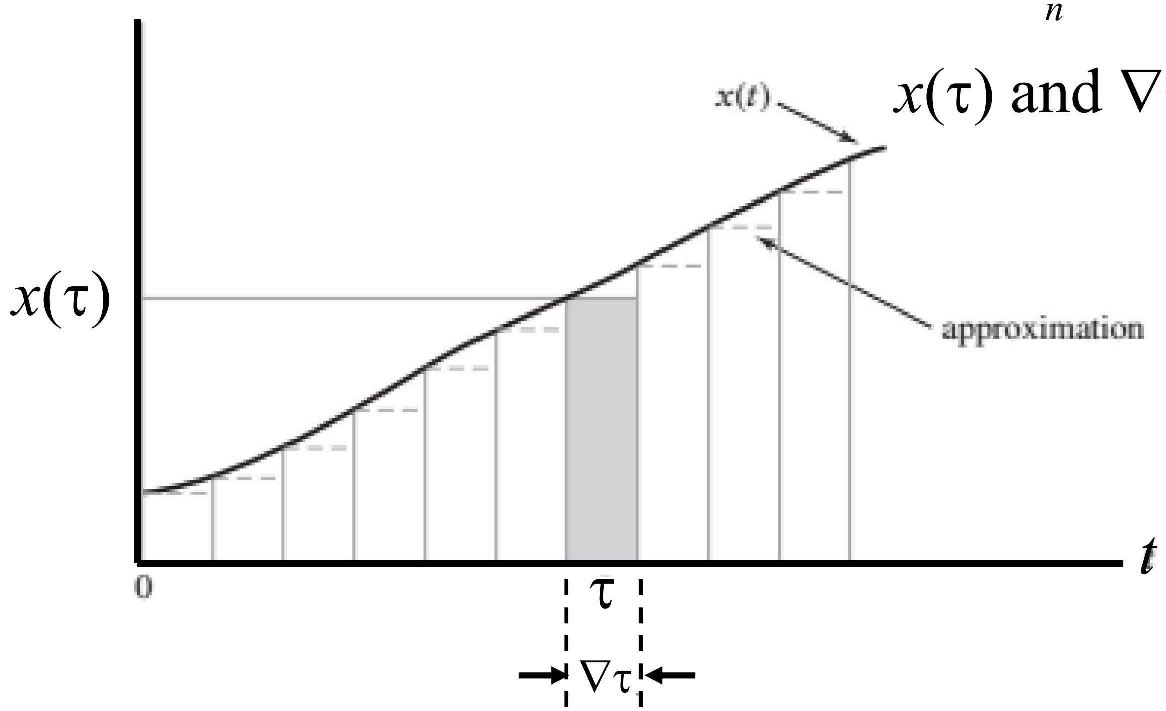

At $t = \tau$, pulse approaches to $\delta(t-\tau)x(\tau)\nabla\tau$

# Signal Transmission through Linear Time Invariant System

Linear Time Invariant System

$$x(t) = \sum_{n} \delta(t - n\nabla\tau)x(n\nabla\tau)\nabla\tau$$

$x(\tau)$ and $\nabla\tau$ are constants




At $t = \tau$, pulse approaches to $\delta(t-\tau)x(\tau)\nabla\tau$

# Signal Transmission through Linear Time Invariant System

Recall Linear Time Invariant System

$$\sum_k a_k \delta(t-t_k) \longrightarrow \boxed{\begin{array}{c}\text{Linear Time}\\ \text{Invariant System,}\\ h(t)\end{array}} \longrightarrow \sum_k a_k h(t-t_k)$$

# Signal Transmission through Linear Time Invariant System

Linear Time Invariant System

$$x(t) = \sum_{n} \delta(t - n\nabla\tau)x(n\nabla\tau)\nabla\tau \longrightarrow \boxed{\begin{array}{c}\text{Linear Time}\\ \text{Invariant System,}\\ h(t)\end{array}} \longrightarrow y(t) = \sum_{n} h(t - n\nabla\tau)x(n\nabla\tau)\nabla\tau$$

# Signal Transmission through Linear Time Invariant System

Linear Time Invariant System

$$x(t)=\sum_{n}\delta(t-n\nabla\tau)x(n\nabla\tau)\nabla\tau \longrightarrow \boxed{\begin{array}{c}\text{Linear Time}\\ \text{Invariant System,}\\ h(t)\end{array}} \longrightarrow y(t)=\sum_{n}h(t-n\nabla\tau)x(n\nabla\tau)\nabla\tau$$

As $\nabla\tau$ approaches zero,

$$y(t)=\sum_{n}h(t-n\nabla\tau)x(n\nabla\tau)\nabla\tau=\sum_{\nabla\tau}h(t-\tau)x(\tau)\nabla\tau$$

# Signal Transmission through Linear Time Invariant System

## Linear Time Invariant System

$$x(t) = \sum_n \delta(t - n\nabla\tau)x(n\nabla\tau)\nabla\tau \longrightarrow \boxed{\begin{array}{c}\text{Linear Time}\\ \text{Invariant System,}\\ h(t)\end{array}} \longrightarrow y(t) = \sum_n h(t - n\nabla\tau)x(n\nabla\tau)\nabla\tau$$

As $\nabla\tau$ approaches zero,

$$y(t) = \sum_n h(t - n\nabla\tau)x(n\nabla\tau)\nabla\tau = \sum_{\nabla\tau} h(t-\tau)x(\tau)\nabla\tau = \int_{-\infty}^{\infty} h(t-\tau)x(\tau)d\tau$$

# Signal Transmission through Linear Time Invariant System

## Linear Time Invariant System

$$x(t) = \sum_n \delta(t - n\nabla\tau)x(n\nabla\tau)\nabla\tau \longrightarrow \boxed{\begin{array}{c}\text{Linear Time}\\ \text{Invariant System,}\\ h(t)\end{array}} \longrightarrow y(t) = \sum_n h(t - \nabla\tau)x(n\nabla\tau)\nabla\tau$$

$y(t) = \int_{-\infty}^{\infty} h(t-\tau)x(\tau)d\tau$ is convolution integral and can be written as

$$y(t) = h(t) * x(t)$$

# Signal Transmission through Linear Time Invariant System

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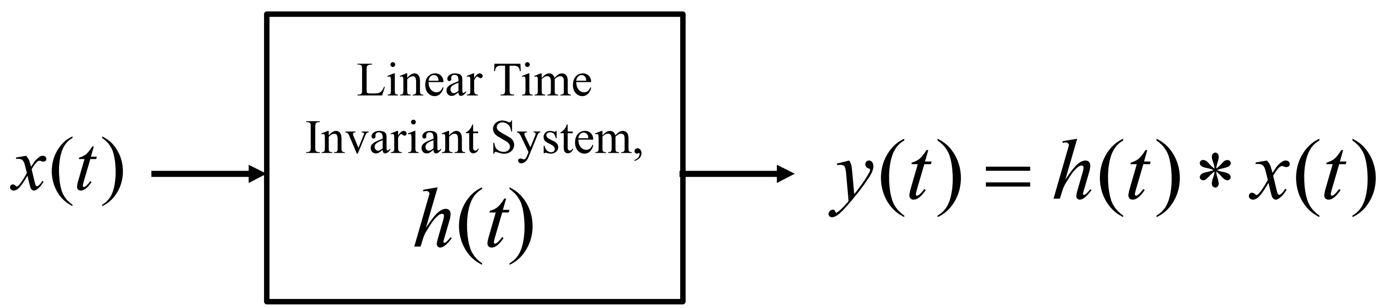
# Signal Transmission through Linear Time Invariant System

Linear Time Invariant System

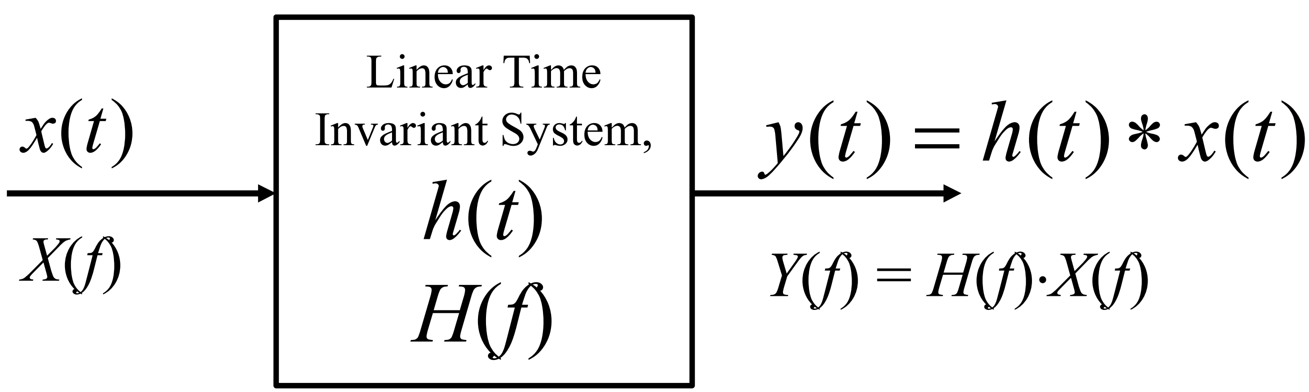



## Now Recall

$$g_1(t) * g_2(t) \Leftrightarrow G_1(f)G_2(f)$$

# Signal Transmission through Linear Time Invariant System

## Linear Time Invariant System

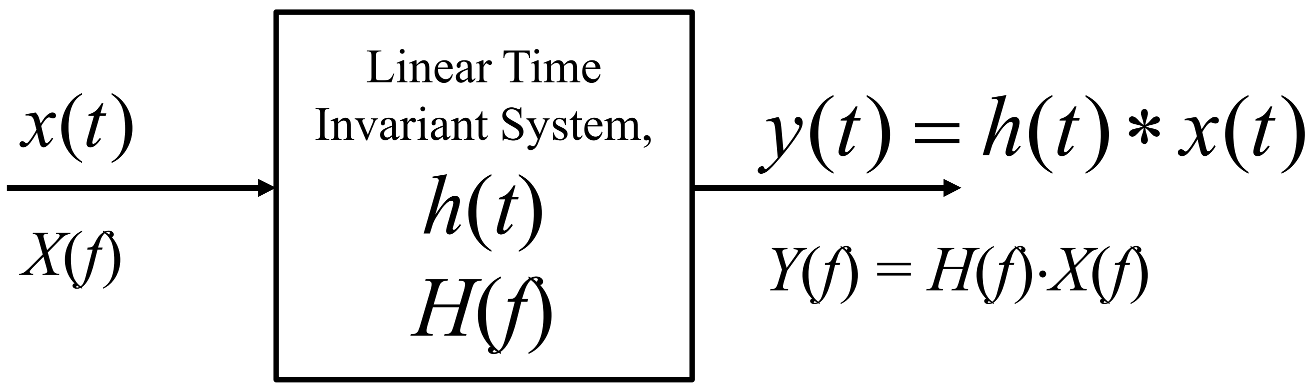



$H(f)$ is

- Fourier transform of $h(t)$
- the frequency response of LTI system
- a transfer function, because it transfers or passes desired frequencies of $X(f)$
- a complex function

# Signal Transmission through Linear Time Invariant System

## Linear Time Invariant System

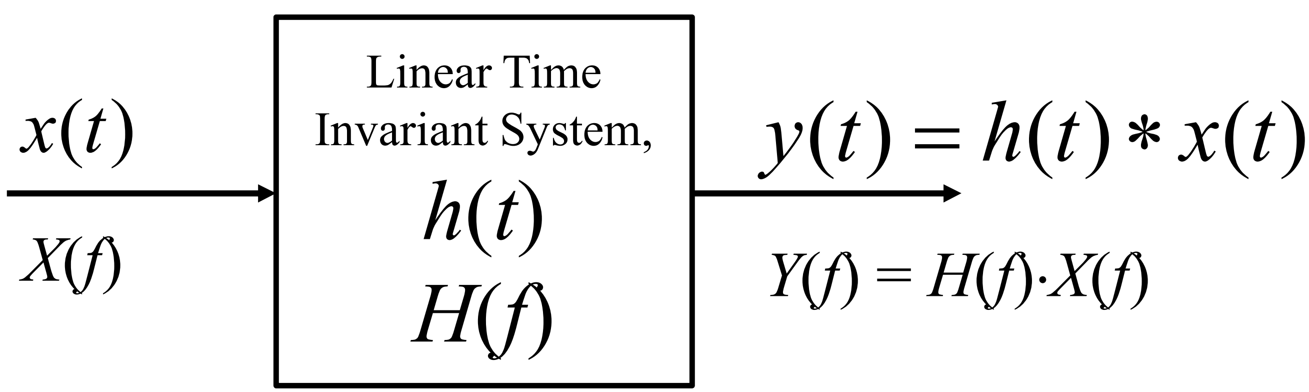



$$H(f) = |H(f)| e^{j\theta_h(f)}$$

- $|H(f)|$ is amplitude response of LTI system
- $\theta_h(f)$ is phase response of LTI system, measured in radians

# Signal Transmission through Linear Time Invariant System

## Linear Time Invariant System



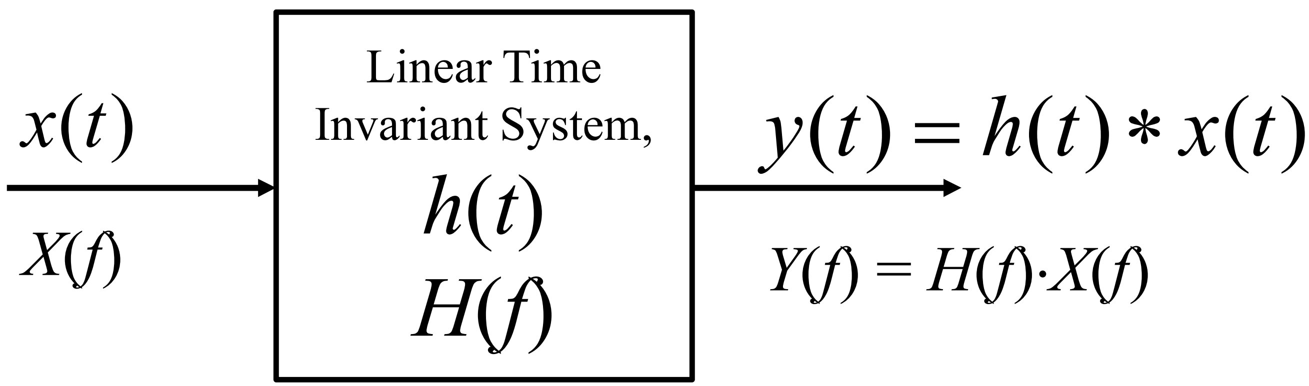

$$H(f) = |H(f)| e^{j\theta_h(f)}$$

$$\ln H(f) = \ln|H(f)| + j\theta_h(f) = \alpha(f) + j\theta_h(f)$$

- $\ln|H(f)|$ or $\alpha(f)$ is the gain of LTI system, measured in nepers

# Signal Transmission through Linear Time Invariant System

## Linear Time Invariant System




$$H(f) = |H(f)| e^{j\theta_h(f)}$$

$$\alpha'(f) = 20 \log_{10} \ln |H(f)| \ \text{dB}$$

- $\alpha'(f)$ is an alternate measurement of gain of LTI system, measured in dB

# Signal Transmission through Linear Time Invariant System

## Causal LTI System

- does not respond before the excitation is applied
- physically realizable system
- impulse response must vanish for negative time

$$h(t)=0, \qquad t<0$$

# Signal Transmission through Linear Time Invariant System

## Causal LTI System

Necessary and sufficient condition for causal system

$$\int_{-\infty}^{\infty} \left( \frac{|\alpha(f)|}{1+f^2} \right) df < \infty \quad \text{where} \quad \alpha(f) = \ln|H(f)|$$

# Signal Transmission through Linear Time Invariant System

## Causal LTI System

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Paley-Wiener Criterion

# Signal Transmission through Linear Time Invariant System

## Causal LTI System

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Paley-Wiener Criterion

### Significance

- If α($f$) satisfy P-W criterion, α($f$) can be combined with $\theta_h(f)$ to find $h(t)$ for a causal system
- Infinite attenuation (α($f$) = -∝ or $|H(f)|$ =0 ) for a band of frequencies DOES NOT satisfy P-W criterion
- However, α($f$) = -∝ allowed for a discrete set of frequencies

# Signal Transmission through Linear Time Invariant System

## Stable LTI System

- output signal is bounded for all bounded input signals.
- The criteria is also known as *bounded input–bounded output* (BIBO) *stability criterion*

# Signal Transmission through Linear Time Invariant System

## Stable LTI System

For bounded input signal, $x(t)$

$$|x(t)| < M \quad \text{for all } t$$

# Signal Transmission through Linear Time Invariant System

## Stable LTI System

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Output from a LTI system

$$y(t) = h(t) * y(t) = \int_{-\infty}^{\infty} h(t-\tau)x(\tau)d\tau = \int_{-\infty}^{\infty} x(t-\tau)h(\tau)d\tau$$

# Signal Transmission through Linear Time Invariant System

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$$|y(t)| = \left| \int_{-\infty}^{\infty} x(t-\tau)h(\tau)d\tau \right| \leq \int_{-\infty}^{\infty} |x(t-\tau)||h(\tau)|d\tau$$

# Signal Transmission through Linear Time Invariant System

## Stable LTI System

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$$|y(t)| = \left|\int_{-\infty}^{\infty} x(t-\tau)h(\tau)d\tau\right| \le \int_{-\infty}^{\infty} |x(t-\tau)||h(\tau)|d\tau \le M \int_{-\infty}^{\infty} |h(\tau)|d\tau$$

# Signal Transmission through Linear Time Invariant System

## Stable LTI System

For bounded input signal, $x(t)$

$$|x(t)| < M \quad \text{for all } t$$

Output from a LTI system

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# Signal Transmission through Linear Time Invariant System

## Stable LTI System

For bounded input signal, $x(t)$

$$|x(t)| < M \quad \text{for all } t$$

Output from a LTI system

$$|y(t)| \leq M \int_{-\infty}^{\infty} |h(\tau)| d\tau$$

ouput signal $y(t)$ to be bounded, $|h(t)|$ must be integrable, that is,

$$\int_{-\infty}^{\infty} |h(\tau)| d\tau < \infty$$

# Signal Transmission through Linear Time Invariant System

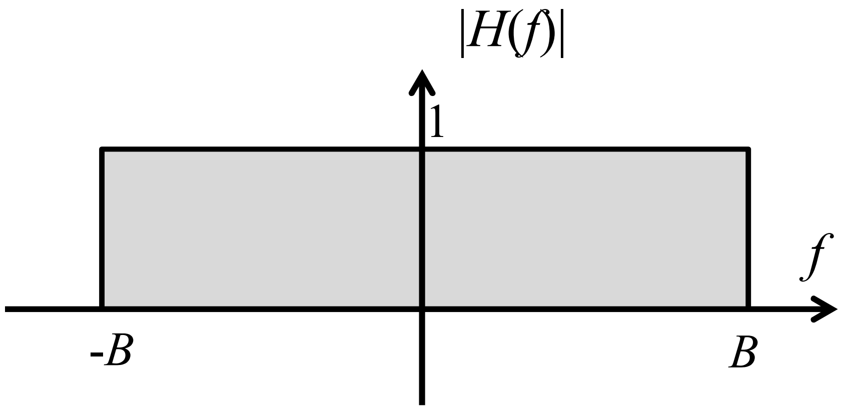
## Stable LTI System

Necessary and sufficient condition for BIBO stability

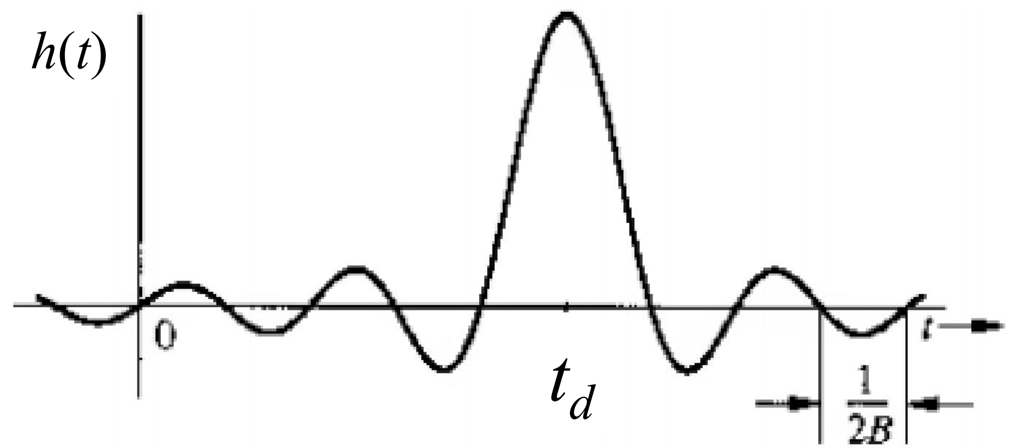
$$\int_{-\infty}^{\infty} \left| h(\tau) \right| d\tau < \infty$$

# Signal Transmission through Linear Time Invariant System

System Response in
Frequency domain




System Response in
Time domain



# Signal Transmission through Linear Time Invariant System

System Response in
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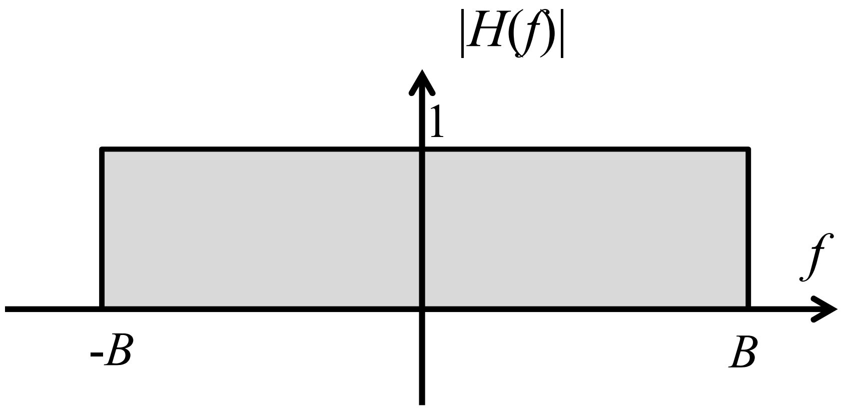
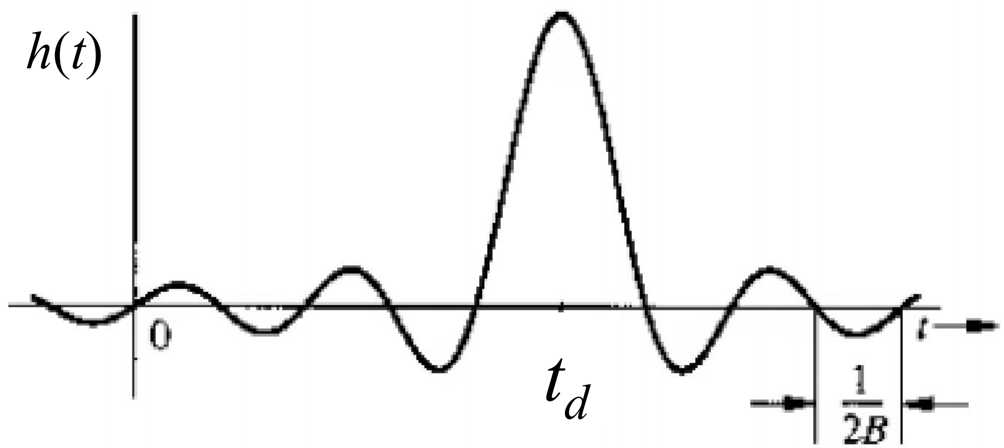
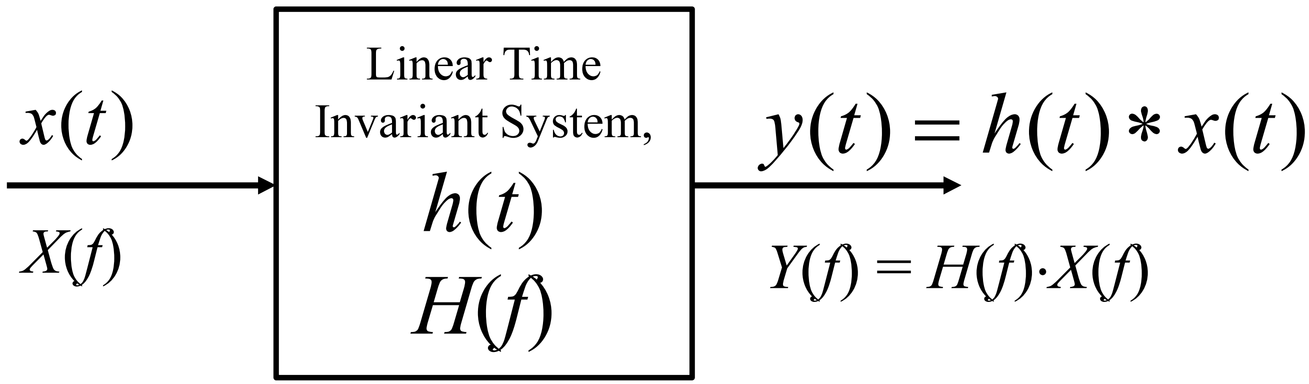



System Response in
Time domain




Is this a causal system?

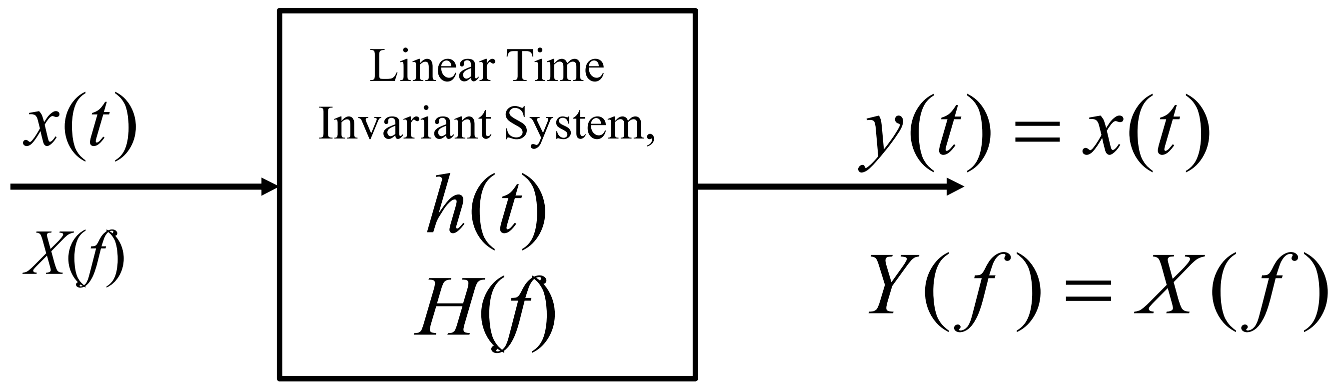
# Signal Distortion during Transmission




$$|Y(f)|e^{j\theta_y(f)} = |H(f)||X(f)|e^{j[\theta_h(f)+\theta_x(f)]}$$

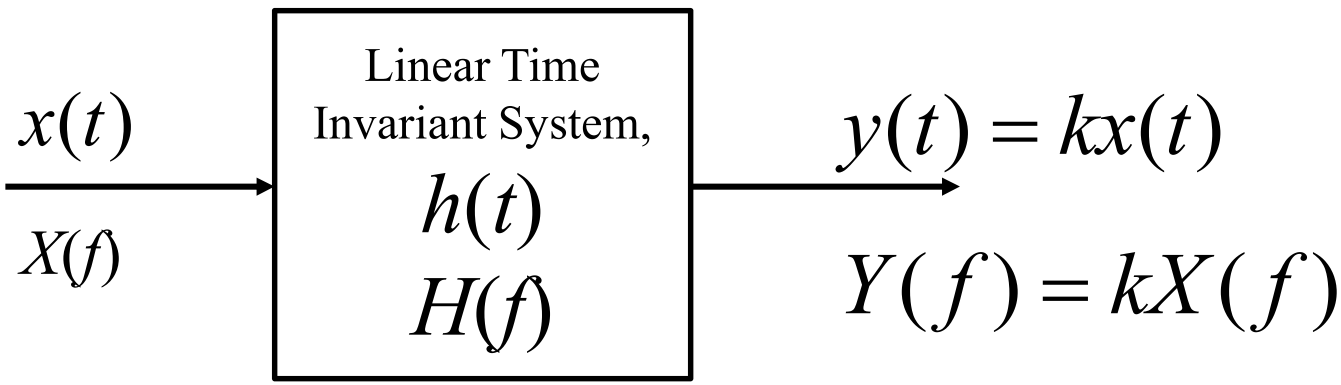
- System distorts the input signal
- Input signal amplitude $|X(f)|$ is changed to $|X(f)|\,|H(f)|$
- Input signal phase is shifted by $\theta_h(f)$
- Plot of $|H(f)|$ and $\theta_h(f)$ vs. $f$ shows how the system changes diffrerent frequency components differently (attenuation/boosting)

# Distortionless Transmission



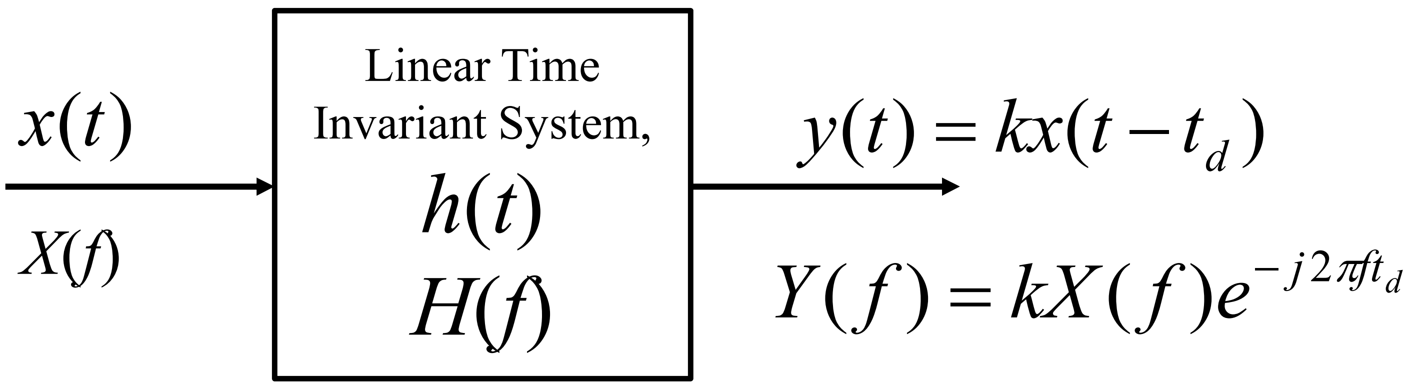

- Input and output signals have identical shape

# Distortionless Transmission



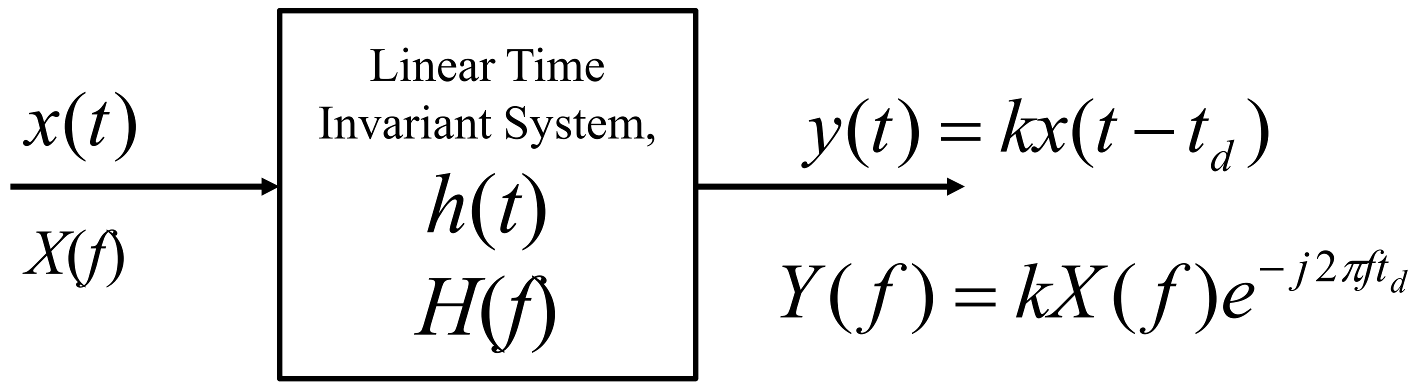

- Input and output signals have identical shape
- Boosting/amplification is allowed

# Distortionless Transmission




- Input and output signals have identical shape
- Boosting/amplification is allowed
- Delaying is allowed

# Distortionless Transmission

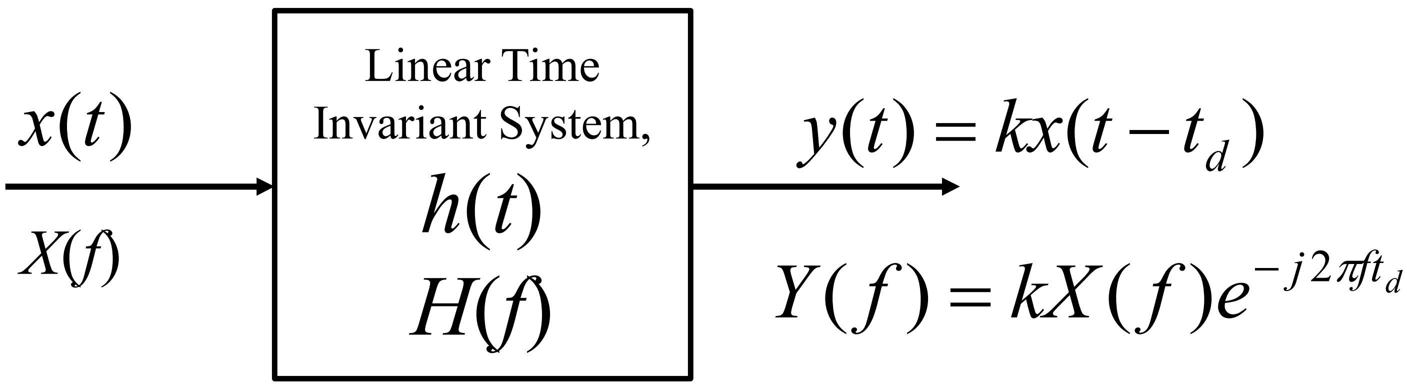



- Input and output signals have identical shape
- Delaying is allowed

$$Y(f) = kX(f)e^{-j2\pi f t_d} = ke^{-j2\pi f t_d} X(f) = H(f)X(f)$$

$$H(f) = ke^{-j2\pi f t_d}$$

# Distortionless Transmission

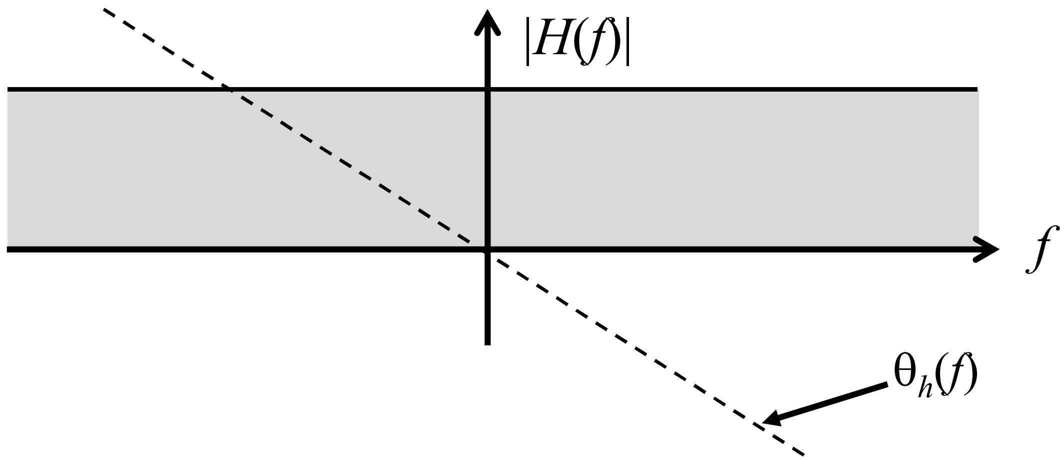



$$H(f) = ke^{-j2\pi f t_d}$$

$$|H(f)| = k \quad \text{and} \quad \theta_h(f) = -2\pi f t_d$$

- $|H(f)|$ is constant
- $\theta_h(f)$ is a linear function of $f$

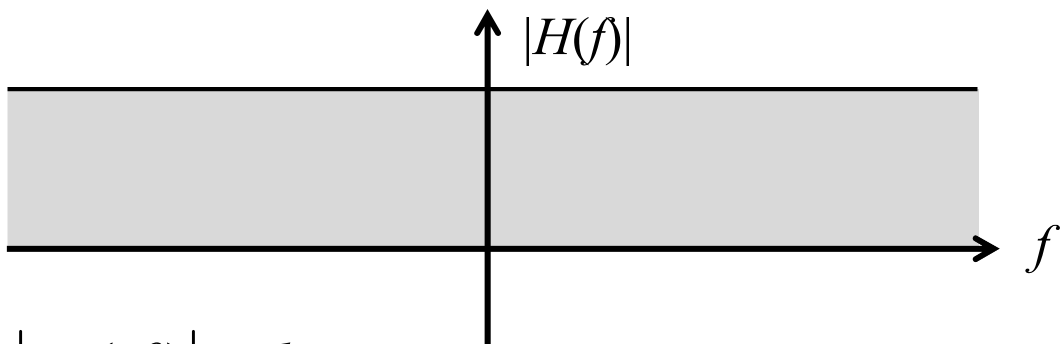
# Distortionless Transmission




$$|H(f)| = k \quad \text{and} \quad \theta_h(f) = -2\pi f t_d = -\omega t_d$$

- $|H(f)|$ is constant
- $\theta_h(f)$ is a linear function of $f$
- Slope of $\theta_h(\omega)$ is $-t_d$, where $t_d$ is delay of the output

# All Pass Vs. Distortionless System

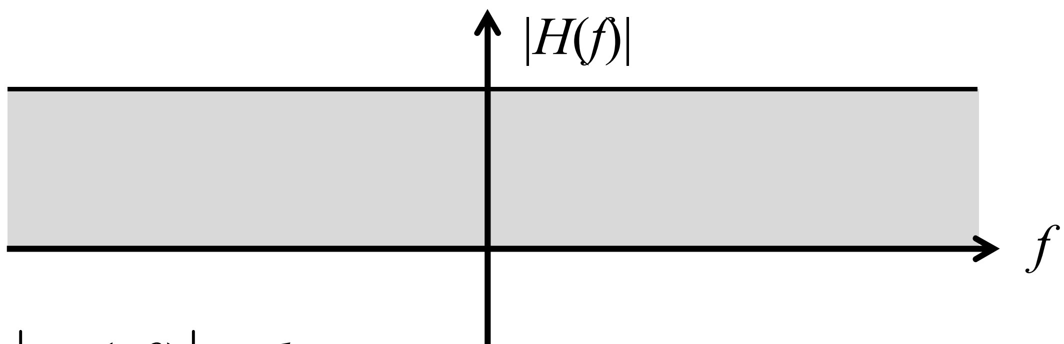



$$|H(f)| = k$$

$$|Y(f)| = |H(f)||X(f)|$$

- $|H(f)| = k$ means that every frequency amplitude of $|X(f)|$ is passed to $|Y(f)|$ with same gain
- Everything is present $|Y(f)|$, nothing is missing!

# All Pass Vs. Distortionless System




$$\left|H(f)\right| = k$$

$$\left|Y(f)\right| = \left|H(f)\right|\left|X(f)\right|$$

- $|H(f)| = k$ means that every frequency amplitude of $|X(f)|$ is passed to $|Y(f)|$ with same gain
- Everything is present $|Y(f)|$, nothing is missing!
- However, this does NOT guarantee distortionless system

# All Pass Vs. Distortionless System

All pass system:

$$\left|H(f)\right| = k$$

Consider a real composite signal consisting of multiple sinusoids of different frequencies

- $|H(f)| = k$ guarantees only presence of all frequency components
- Different sinusoid may get different delays.
- Synchronization of multiple frequency components is NOT guaranteed
- Output composite signal will be out of sync

# All Pass Vs. Distortionless System

Distortionless system:

$$|H(f)| = k \quad \text{and} \quad \theta_h(f) = -2\pi f t_d$$

- Linear phase $\theta_h(f) = -2\pi f t_d$ guarantees a constant time delay $t_d$ for every sinusoid
- Synchronization is guaranteed

# All Pass Vs. Distortionless System

How to Check Distortionless transmission?

Find the slope (derivative) of $\theta_h(f)$

$$t_d(f) = -\frac{1}{2\pi} \cdot \frac{d\theta_h(f)}{df}$$

# All Pass Vs. Distortionless System

How to Check Distortionless transmission?

Find the slope (derivative) of $\theta_h(f)$

$$t_d(f) = -\frac{1}{2\pi} \cdot \frac{d\theta_h(f)}{df}$$

Hint: Rearrange and differentiate $\theta_h(f) = -2\pi f t_d$